

BUCK CONVERTER

The schematic diagram illustrates a Buck Converter circuit. The input is a 24V_BUCK supply connected to the VIN pin (pin 4) of the IC1. A 2.2uF/100V capacitor (C6) is connected between the input and ground (GND2). A 0.1uF/X7R capacitor (C5) is connected between the VIN pin and the EN_UVLO pin (pin 3). The EN_UVLO pin is also connected to the VCC pin (pin 7). The VCC pin is connected to the MODE_SYNC pin (pin 1) and the PGOOD pin (pin 2). The MODE_SYNC pin is connected to ground (GND2) through a 1uF/X7R capacitor (C9). The PGOOD pin is connected to ground (GND2). The BOOT pin (pin 6) is connected to the SW pin (pin 5) through a 0.1uF/X7R capacitor (C1). The SW pin is connected to the inductor L1 (4.7uH). The other end of the inductor is connected to the output of the converter. The output is connected to a 100Ohm resistor (R1) and a 43.2K resistor (R2) in series, which is connected to ground (GND2). The output is also connected to a 22uF/16V/X7R capacitor (C2) and a 22uF/16V/X7R capacitor (C3) in parallel, which is connected to ground (GND2). The output voltage is labeled 3.3V_DI.

ISOLATED DC/DC

The diagram illustrates an isolated DC/DC converter circuit. The main circuit (PS1) is a transformer-coupled converter with a 22uH / 99.6mOhm inductor (L2) and a 10uF / 1210 capacitor (C17) on the input. The output is connected to a 3.3V MCU_UNREG rail. A 3.3V Fixed LDO (IC6) is used to regulate the output to 3.3V MCU. The LDO has a 1uF / X7R capacitor (C16) on its input and a 1uF / X7R capacitor (C8) on its output. The LDO's output is connected to a 3.3V MCU rail. The LDO's input is connected to the 3.3V MCU_UNREG rail. The LDO's output is connected to a 3.3V MCU rail. The LDO's output is connected to a 3.3V MCU rail.

- ☐ DIRTY
- ☐ CLEAN
- ☒ ISOLATION

The schematic diagram illustrates a 24V Buck converter circuit. The components are organized into four distinct functional blocks, each enclosed in a dashed box:

- TVS (Transient Voltage Suppressor):** Located at the input, it consists of a TVS diode (D1) connected to the 24V input line (pin 1) and ground (pin 2, labeled GND1).
- P-FET (Power Field-Effect Transistor):** The switching element, represented by a P-Channel MOSFET (Q1). Its gate is driven by a PTZTFTE2515B MOSFET driver (Z1) through a 10kOhm resistor (R3). The drain is connected to the 24V input line (pin 2), and the source is connected to ground (pin 1, labeled GND1).
- CMC (Common Mode Choke):** A common mode choke (FL1) is placed in series with the output line (pin 1) to filter out common mode noise. It is connected between the P-FET drain and the buck converter input.
- CLC (CLC Filter):** A CLC filter (FL2) is used for output filtering. It is connected between the CMC output and the buck converter input. The buck converter output (pin 2, labeled 24V BUCK) is connected to a load, which is represented by a 100nF/100V capacitor (C11).

The circuit also includes several other components and connections:

- A 24V input line (pin 1) and a 24V output line (pin 2, labeled 24V BUCK).
- A 100nF/100V capacitor (C4) connected between the CMC output and ground (pin 2, labeled GND2).
- A 100nF/100V capacitor (C10) connected between the buck converter input and ground (pin 2, labeled GND2).
- A 100nF/100V capacitor (C11) connected between the buck converter output and ground (pin 2, labeled GND2).
- A 10kOhm resistor (R3) connected between the MOSFET driver (Z1) and the P-FET gate.
- A PTZTFTE2515B MOSFET driver (Z1) connected between the P-FET gate and ground (pin 1, labeled GND1).

[illegible]

DEBUG

	J2
SWCLK	1
SWSDIO	2
GND3	3
XRES	4
3.3V MCU	5

Male-1x5

MOLEX

J1

1	KL30
2	KL15 IG
3	GND1
4	CAN P
5	CAN N

Male-1x5

Automotive CAN-to-Bluetooth Interface		
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